→ Disadvantages of In memory

→ Types of Persistent Storage (High Freq. Trader {Goldman, Tower})

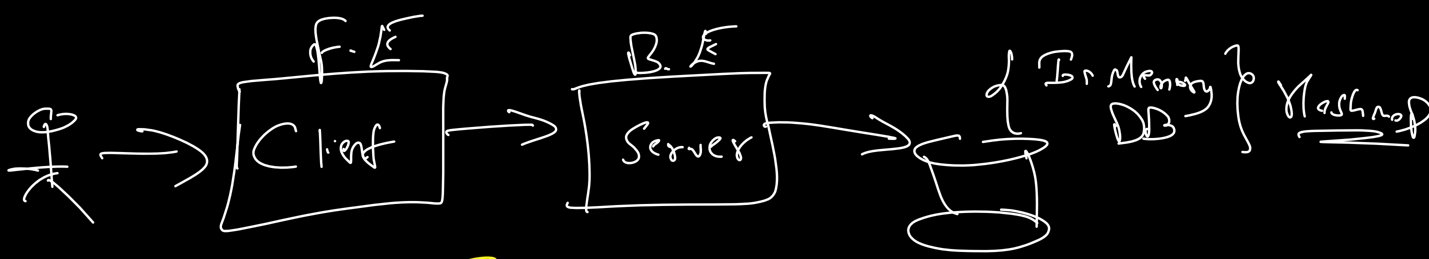
→ Load Balancer

→ Relational / NoSql

→ CAP {Consistency, Availability, Partition Tolerance}

→ Book Management

→ Lombok, Validation, Builder



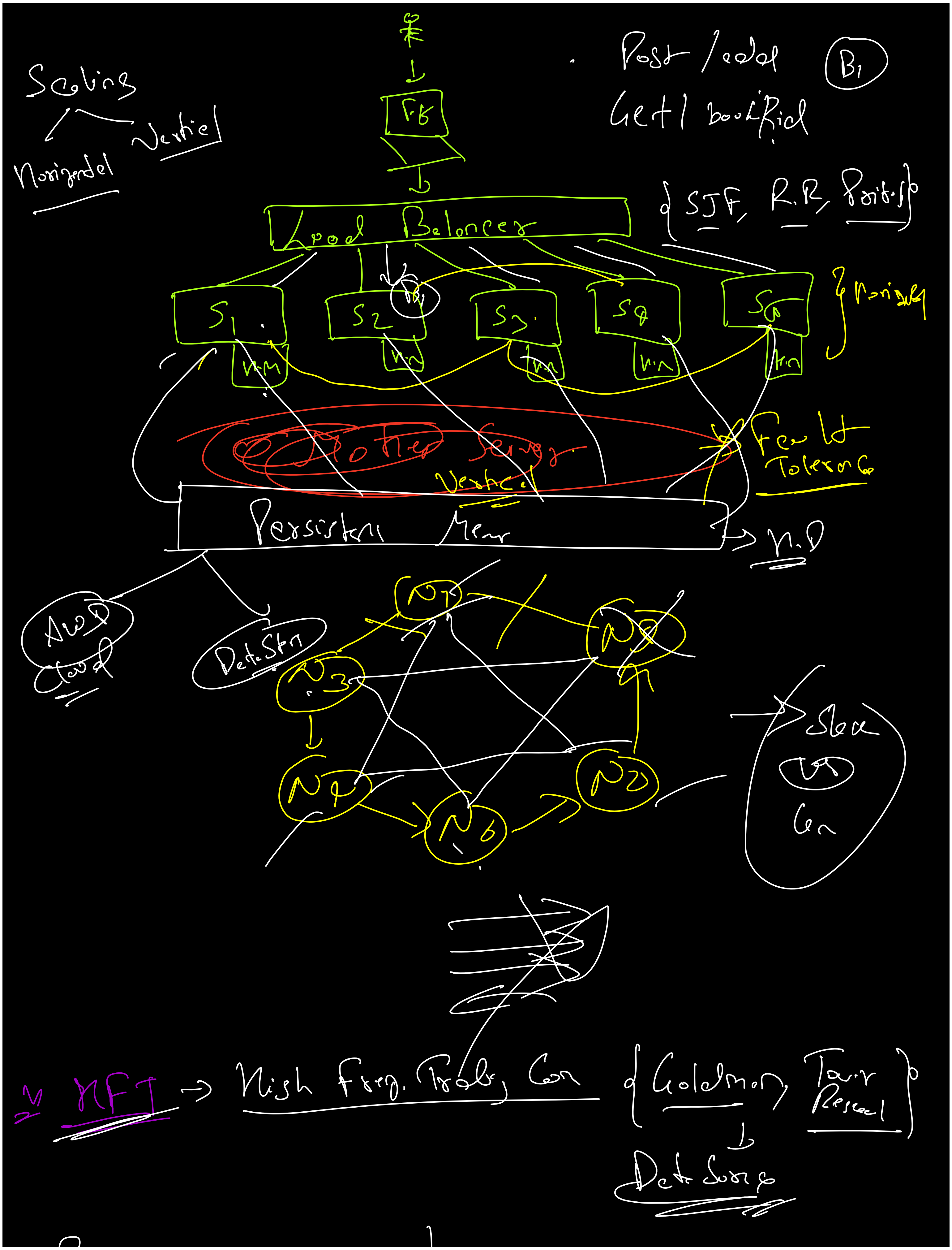

* Pros & Cons of In Memory

→ Cons
i) Volatile
ii) Limit on Data Size
iii) RAM {limited}

Pros
i) Faster Access

Max Disk 1TB → 5000–7000 → 8 GB → HDD → ROM
SSD


→ Amazon → Month → Music, Del., Prime } 10–50M
→ Day → 1 Million → 1sec 6→7 lakh


Scaling
Horizontal
Vertical
Post /add
Get / book?id
B1
{SJF, R.R, Priority}
FE
Load Balancer
S1
S2
S3
S4
S5
V.M
Mother Server
Vertical
Fault Tolerance
Persistent Mem
N.D
AWS Cloud
Data Store
N1
N2
N3
N4
N5
N6
N7
Slack
Go
NFT → High Freq. Trading Com {Goldman, Tower Research}
Data Store

| Persist | In Mem |
| --- | --- |
| 1.) Fault Tolerance | 1.) No fault Tol |
| 2.) ↓ Speed | 2.) ↑ Speed |
| 3.) non Volatile | 3.) Volatile. |
| 4.) ↑ Size | 4.) ↓ Size |
| 5.) ↓ Cost | 5.) ↑ Cost |

* MySQL →

Relational
→ MySQL, Postgres, Oracle
ACID

NoSql
→ Mongo, JSON

A → Atomicity → All or None.
C → Consistency → Before / After db opn.
I → Isolation → Transactions should be isolated → RR, → RW, → WR, → WW
D → Durability →

* Distributed Syst

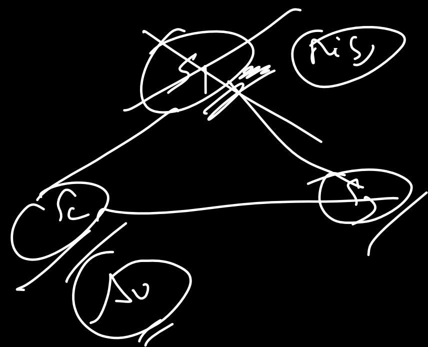
CAP theorem → Consistency
→ Availability
→ Partition

→ A → foo
B → foo

Consist
No avail

Youtube / Insta Clicked → User stop like

Then cancel like



Copy

State

select * from x where x = 1;

C → S → DB → " " " " x = 2;

DB → Cache → Complete

→ Run

$C_1 + R_1$ & $C_2 + R_2$

$2C + 2R$

$C_1$ $R_1$

select * from x where x = 2

x = 1, x = 2